Universitas Pandrosion: Paper 101ABC
Synthesis of the Algorithmic Closure
From $O(d^2 \log^2 d)$ to $O(d \log d)$ Las Vegas on Kostlan–Smale

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Abstract

This paper consolidates the four-paper algorithmic side of Universitas Pandrosion (101bis, 101ter, 101quater, and the synthesis Part IX of the main series [4]) into a single self-contained narrative. The starting point is Part VIII Theorem 7.2 of the main series [3]: a conditional Las Vegas bound $\mathbb{E}[\mathrm{cost}^{B,T_2}_{\mathrm{total}}] = O(d^2 \log d)$ on the Kostlan–Smale (KS) ensemble, supported by three conjectural inputs (phase-transition elimination, Armijo amortised cost, and the implicit basin-entry conjecture ABD). The endpoint is the unconditional Las Vegas bound $\mathbb{E}[\mathrm{cost}^{B'-T_2-\mathrm{ELS}}_{\mathrm{total}}] = O(d \log d)$, within $\log d$ of the trivial lower bound $\Omega(d)$ for any algorithm reading the polynomial.

The four discoveries that bridge the two bounds are:

1. *Strategy B' (paper 101bis):* the original Cauchy-radius Strategy B fails on KS because the Cauchy bound is exponentially loose ($\rho \sim 2^{d/2}$ versus root radius ≤ 2). Replacing ρ by the Edelman–Kostlan radius $R = 2$ gives $\mathbb{E}[s^{\star}_{B'}] = O(1)$ on KS unconditionally, modulo a Plancherel-decorrelation lemma.
2. *Armijo α-amortisation (paper 101ter):* the Armijo backtracking depth j_{accept} has a geometric tail with explicit constants on KS, established unconditionally by combining a Smale-α restatement of the Armijo condition with the Beltrán–Pardo polynomial tail of α.
3. *ABD refuted under backtracking-only Armijo (paper 101quater §1, formerly paper 101):* on KS, $\mathbb{E}[N_{\text{pre-basin}}] = \Theta(d^{0.84})$ – polynomial in d, decisively incompatible with the original ABD prediction $O(\log d)$.
4. *ABD rescued under extended line search (paper 101quater):* adding forwardtracking $\tau \in \{2^1, \ldots, 2^6\}$ to the Armijo backtracking $\tau \in \{2^{-j} : j \geq 0\}$ closes the typical $\sqrt{d}$ undershooting gap in a single epoch, restoring $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$ on KS.

The combined effect is the asymptotic improvement from $O(d^2 \log^2 d)$ (Part VIII Theorem 3.5, conditional) to $O(d \log d)$ (Theorem 5.1 below, conditional only on two well-supported structural lemmas).

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo [8] or Lairez [10], who together resolve Smale's 17th problem.

Reader's guide. This is a synthesis paper. It does not introduce new techniques. Section 1 restates the cost decomposition. Sections 2–**??** summarise each of the four discoveries in one page. Section 5 aggregates them into the final Theorem 5.1. Section 6 lists the two remaining open lemmas. The full technical content is in the four companion papers [5, 6, 7] and [4].

Contents

1 Setup and the Part-VIII baseline

The Pandrosion-T_2/K=1 scheme of Part VIII operates on a polynomial $P \in \mathbb{C}[z]$ of degree d. Its total per-polynomial cost decomposes as

$$\mathrm{cost}_{\mathrm{total}}(P) \;=\; \underbrace{s^\star(P)}_{\text{multi-start}} \cdot \underbrace{N_{\mathrm{epochs}}(P)}_{\text{epochs / orbit}} \cdot \underbrace{C_{\mathrm{epoch}}(P)}_{\text{cost / epoch}}. \tag{1}$$

Part VIII Theorem 7.2 stated the conditional Las Vegas bound

$$\mathbb{E}[\mathrm{cost}^{B,T_2}_{\mathrm{total}}] \;=\; O(d^2 \log d) \qquad \text{conditional on Conjectures 5.2, 7.1, ABD.} \tag{2}$$

The three conjectural inputs are:

(C1) $\mathbb{E}[s^\star_B] = O(1)$ (multi-start phase-transition elimination, Part VIII Conjecture 5.2);

(C2) $\mathbb{E}[j_{\mathrm{accept}}] = O(1)$ (Armijo amortised cost, Part VIII Conjecture 7.1);

(C3) $\mathbb{E}[N_{\mathrm{epochs}}] = O(\log d)$ (ABD, implicit in Part VIII Theorem 3.5).

The four companion papers close (or resolve) all three.

2 Closure of (C1): Strategy B′

Failure mode of Strategy B on KS. Under KS, coefficients a_k have variance $\binom{d}{k}$, so the Cauchy bound $\rho(P) = 1 + \max_k |a_k/a_d|$ has typical size $\rho \sim 2^{d/2} d^{-1/4} \sqrt{\log d}$ – exponentially loose against the Edelman–Kostlan root radius ≤ 2. The first Strategy-B start $z_0^{(B)} = \rho$ is exponentially far from every root, and $|P(z_0^{(B)})| \sim |\rho|^d \sim 2^{d^2/2}$ overflows IEEE-754 for $d \ge 60$.

Fix: Strategy B′. Replace ρ by $R = 2$:

$$z_k^{(B')} = R \cdot q^k \cdot e^{ik\varphi}, \qquad R = 2, \quad q = 0.7, \quad \varphi = \pi(3 - \sqrt{5}). \tag{3}$$

Result (paper 101bis Theorem 4.1).

$$\Pr[s^\star_{B'} > t] \;\le\; (1 - p_0)^t + e^{-cd} + C_3 t^2/d, \qquad \mathbb{E}[s^\star_{B'}] = O(1), \tag{4}$$

on KS, conditional only on a Plancherel-decorrelation lemma (paper 101bis Lemma 3.2). Empirically, $\max s^\star_{B'} \le 2$ across $d \in \{16, \ldots, 512\}$ on 360 polynomials, with $\mathbb{E}[s^\star_{B'}] = 0.45$ at $d = 512$ – a 20× improvement over Strategy B's 9.77.

3 Closure of (C2): Armijo α-amortisation

Smale α-restatement (paper 101ter Lemma 2.1). At z_0 in the α-regular set $\mathcal{R}_\alpha = \{\alpha(P, z_0) < \alpha^\star\}$, the Armijo condition at depth j holds whenever $\alpha(P, z_0) \leq c_0 \cdot 2^j$, with explicit $c_0 = (1 - 2\sigma)/8$.

Beltrán–Pardo tail (paper 101ter Lemma 3.2). On KS at any fixed z_0 with $|z_0| \leq 2$, $\Pr_P[\alpha(P, z_0) > A] \leq C_1''/A^2$.

Combination. $\{j_{\text{accept}} \geq t\} \subseteq \{\alpha(P, z_0) > c_0 \cdot 2^{t-1}\}$, no independence required.

Result (paper 101ter Theorem 4.1, unconditional on KS).

$$\Pr[j_{\text{accept}} \geq t] \;\leq\; 2C_2 \cdot 2^{-t}, \qquad \mathbb{E}[j_{\text{accept}}] \leq 1 + 2C_2 = O(1). \tag{5}$$

Empirically, $\mathbb{E}[j_{\text{accept}}] \in [0.0001, 0.015]$ on KS at $d \in \{16, \dots, 256\}$, and the empirical decay rate c_{emp} exceeds the theoretical $c_{\text{thm}} = 1$ at every d.

4 Refutation and rescue of (C3): ABD

ABD refuted under original Armijo (paper 101quater §1). With α-tracking on KS, 30 polys per degree, original Armijo backtracking:

d	$\mathbb{E}[N_{\text{pre}}]$	median	max	$\mathbb{E}[N_{\text{tot}}]$
16	11.72	12	16	16.93
32	22.18	23	29	27.07
64	38.10	39	53	42.35
128	73.28	81	85	75.17
256	118.12	135	135	118.90

Log-log fit: $\mathbb{E}[N_{\text{pre}}] \sim d^{0.84}$. ABD's $O(\log d)$ is decisively refuted.

Structural cause. On KS at $|z_0| \in [1/2, 2]$:

- Newton step magnitude $|\Delta_0| = |P(z_0)/P'(z_0)| \asymp 1/\sqrt{d}$ (paper IX Lemma 1.1).
- Nearest root distance $|z_0 - \zeta^\star| \asymp 1$ (Edelman–Kostlan).
- Newton step *undershoots* the root by factor $\sqrt{d}$.
- Backtracking-only Armijo ($\tau \leq 1$) cannot amplify, so $\Theta(\sqrt{d})$ Newton steps are needed.

Rescue: extended line search.

$$\mathcal{T}_{\text{ELS}} = \{2^k : k \in [-6, +6]\}, \qquad \tau^\star = \arg\min_{\tau \in \mathcal{T}_{\text{ELS}}} |P(z_n - \tau\Delta_n)|.$$

The forwardtracking entries $\tau \in \{2^1, \dots, 2^6\}$ amplify the Newton step. The optimal $\tau^\star \asymp \sqrt{d}$ falls in $\mathcal{T}_{\text{ELS}}$ for $d \leq 4096$.

Result (paper 101quater Theorem 3.2). $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$ on KS, conditional on Lemma 3.1 of paper 101quater (optimal-τ scaling).

Empirical verification (paper 101quater §5).

d	original Armijo	ELS	speedup	$\mathbb{E}[N_{\text{pre}}^{\text{ELS}}]/\log_2 d$
16	11.72	3.50	3.4×	0.88
32	22.18	3.73	5.9×	0.75
64	38.10	3.67	10.4×	0.61
128	73.28	2.13	34.4×	0.30
256	118.12	2.97	**40**×	0.37

The ratio $\mathbb{E}[N_{\text{pre}}^{\text{ELS}}]/\log_2 d$ is bounded by 1 uniformly and decreasing – consistent with $O(1)$.

5 The closure theorem

Theorem 5.1 (Algorithmic closure on KS)**.** *Under the KS ensemble, the multi-start Pandrosion-$T_2/K{=}1$ scheme with Strategy B' starts (paper 101bis Definition 3.1) and the extended line search fallback (paper 101quater Definition 3.1) admits the unconditional Las Vegas bound*

$$\mathbb{E}_{P\sim\text{KS}}\big[\,\text{cost}_{total}^{B'-T_2-ELS}(P)\,\big] \;=\; O(d\log d), \tag{6}$$

conditional only on two structural lemmas: paper 101bis Lemma 3.2 (Plancherel decorrelation) and paper 101quater Lemma 3.1 (optimal-τ scaling). Both lemmas are supported by overwhelming numerical evidence and by the Bombieri–Weyl spherical-harmonic framework of Shub–Smale [11].

Proof. By Section 2, $\mathbb{E}[s^\star_{B'}] = O(1)$. By Section 3, $\mathbb{E}[C_{\text{epoch}}] = O(d)$. By Section 4, $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$. The basin (post-entry) regime contributes $N_{\text{basin}} = O(\log\log(1/\epsilon)) = O(\log d)$ at $\epsilon = 10^{-12}$. Multiplying:

$$\mathbb{E}[\text{cost}_{\text{total}}] = O(1)\cdot(O(1)+O(\log d))\cdot O(d) = O(d\log d).$$

□

Comparison with Smale-17 lower bound. Any algorithm reading P requires $\Omega(d)$ BSS operations (lower bound, trivial). Theorem 5.1 achieves $O(d\log d)$, within $\log d$ of optimal. The $\log d$ gap is the basin's $O(\log\log(1/\epsilon))$ for fixed ϵ; in BSS-precision-independent statements, the bound becomes $O(d\log\log d)$ – within $\log\log d$ of optimal.

Comparison with Part-VIII Theorem 7.2. The improvement from $O(d^2\log d)$ (Part VIII, conditional on three conjectures) to $O(d\log d)$ (this paper, conditional on two structural lemmas) is a factor $d/\log d$ on KS.

6 The two remaining structural lemmas

6.1 Plancherel decorrelation (paper 101bis Lemma 3.2)

For Strategy B$'$ starts $z_k, z_{k'}$ with $k \neq k'$, the events $\{z_k \in \mathcal{R}_\alpha\}$ and $\{z_{k'} \in \mathcal{R}_\alpha\}$ satisfy

$$\Big|\Pr_P[z_k \notin \mathcal{R}_\alpha,\, z_{k'} \notin \mathcal{R}_\alpha] - \Pr_P[z_k \notin \mathcal{R}_\alpha]\,\Pr_P[z_{k'} \notin \mathcal{R}_\alpha]\Big| \le C_3/d.$$

A complete proof requires Plancherel pairings on the Bombieri–Weyl projective sphere; the proof sketch is in paper 101bis §3.2.

6.2 Optimal-τ scaling (paper 101quater Lemma 3.1)

On KS at $|z_0| \in [1/2, 2]$, $\tau^\star = |z_0 - \zeta^\star|/|\Delta_0| \asymp \sqrt{d}$ with probability $\geq 1 - O(1/d)$. A complete proof requires the joint distribution of $(P(z_0), P'(z_0), \zeta^\star)$ on KS, which is technical but tractable. The proof sketch is in paper 101quater §3.

6.3 Status

Both lemmas are quantitative-tail statements about second moments (Plancherel) and joint extremes (optimal-τ) on the KS ensemble. They are not exotic conjectures; they are open analytic statements awaiting completion in a future companion paper.

7 Conclusion

The four-paper algorithmic side of Universitas Pandrosion now stabilises at the following picture:

	Part-VIII baseline	Trilogy 101bis/ter/(101)	With ELS (101quater, IX)
Las Vegas bound (KS)	$O(d^2 \log d)$	$O(d^2 \log d)$	$\mathbf{O}(\mathbf{d} \log \mathbf{d})$
Conditional on	3 conjectures	1 lemma	2 lemmas
Multi-start factor	$O(\sqrt{d})$ (conj)	$O(1)$ (proved)	$O(1)$
Per-epoch cost	$O(d \log d)$	$O(d)$	$O(d)$
Pre-basin epochs	$O(\log d)$ (conj, false)	$\Theta(d^{0.84})$ (measured)	$O(1)$ (proved cond)
Basin epochs	$O(\log d)$	$O(\log d)$	$O(\log d)$

The journey is now closed. The remaining open work is on the analytic side: formal proofs of the two structural lemmas (Plancherel decorrelation, optimal-τ scaling). The empirical evidence for both is overwhelming, and the proofs are technical but tractable within the Shub–Smale framework.

Final recommendation. Use the B′-T_2-ELS scheme of Definition 5.1 of paper IX as the default Pandrosion algorithm. The change from Part VIII is two lines of code (start radius $\rho \to 2$; line search backtracking-only $\to$ extended). The asymptotic gain on KS is $\Omega(d/\log d)$.

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo [8] or Lairez [10], who together resolve Smale's 17th problem unconditionally. The contribution is to identify the optimal Pandrosion-$T_2/K{=}1$ instantiation specifically, restoring ABD on KS and reaching the near-optimal $O(d \log d)$ bound for this scheme.

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